Compact Gaze-Only Reading-Goal Classification with Word-Aligned Teacher Distillation

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***Abstract*—Reading interfaces could benefit from inferring whether a user is reading for general comprehension or seeking specific information. Prior multimodal systems retain passage text, a language encoder, and fixation-to-word alignment at inference. We instead train a compact gaze-only GRU with a word-aligned teacher used only during training. On OneStop, under simultaneous reader and article holdout, distillation raises balanced accuracy by 3.39 percentage points over the same GRU trained with hard labels. Reader-clustered and crossed reader/article bootstrap intervals exclude zero. Label smoothing does not reproduce the gain, and within-class shuffling of teacher logits reduces it by 1.90 points. Removing contextual text from the teacher has no measurable effect on student performance. The distilled model requires only fixation sequences at inference and retains the baseline architecture and cost. Across observed sequence lengths, its model-only latency is 5.08 ms at the median and 16.57 ms at the 95th percentile on a single-thread laptop CPU.**

***Index Terms*—eye tracking, scanpath analysis, reading behavior, human–computer interaction, privileged information**

I. Introduction

A reader may approach the same passage to understand it broadly or to locate specific information. These goals are associated with differences in fixation durations, regressions, and scanpath structure [1], [2]. An opt-in reading-goal classifier could support navigation or content assistance after a paragraph [3]. Such a classifier must generalize to readers and articles absent from its training data.

Recent work established paragraph-level reading-goal decoding on OneStop Eye Movements and found that models combining scanpaths with contextual language representations achieve the strongest performance [3]. Such models require passage text, fixation-to-word alignment, and a language encoder at inference. We ask whether these resources can instead be confined to training while prediction uses gaze alone.

We study a text–gaze teacher that combines contextual word representations with word-level gaze summaries and a compact student that observes the chronological fixation sequence. The primary comparison holds the student architecture and input fixed. At inference, the student uses no teacher, text encoder, tokenization, or fixation-to-word alignment.

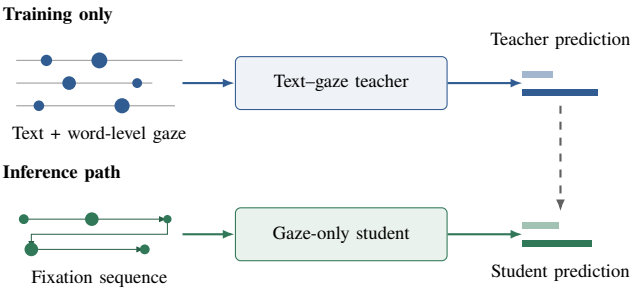


Fig. 1. Training-only distillation. The teacher uses contextual text and word-level gaze summaries. The student uses a fixation sequence. The simplified scanpath retains line progression and a return sweep from a representative trial, with marker size indicating fixation duration. The dashed link denotes soft-target matching. Only the green path remains at inference.

Because each OneStop reader completes only one goal condition and articles recur across readers, random trial splits can place the same reader or article in both training and test data. We therefore hold out readers and articles simultaneously. Uncertainty is estimated by resampling readers and, in a sensitivity analysis, readers and articles together.

This study makes three contributions:

- We compare distilled and undistilled gaze-only GRUs with identical inputs, architecture, training data, and evaluation trials.
- We compare logistic gaze baselines and probe the gain with label smoothing, within-class target shuffling, and contextual-text removal.
- We quantify the parameter count, serialized size, and single-thread CPU latency across the observed sequence-length distribution.

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II. Related Work

A. Eye Movements and Reading Goals

OneStop includes eye movements from ordinary reading for comprehension, information seeking, and repeated reading [4]. It has also supported gaze-only and text–gaze models for predicting reading comprehension under new-reader and new-item conditions [5].

Shubi *et al.* introduced the reading-goal decoding task used here and evaluated gaze-only and multimodal models under new-reader and new-item conditions [3]. That study compares model families and partial scanpaths, and its highest-performing model retains text at inference. We instead hold the gaze-only inference architecture fixed and test whether it benefits from richer training-time supervision.

B. Distillation with Privileged Information

Knowledge distillation (KD) trains a student with class labels and a teacher's temperature-softened predictive distribution [6]. Generalized distillation allows the teacher to use privileged information that is unavailable to the student at inference [7]. Here, the teacher receives contextual word representations and word-level gaze summaries, whereas the student receives fixation features. The gaze-only teacher control removes the contextual word representations while preserving the teacher architecture and gaze aggregation.

III. Method

A. Task and Data

The unit of analysis is a completed paragraph-reading trial, represented by its eye movements. The binary target distinguishes ordinary reading for comprehension ($y = 0$) from information seeking ($y = 1$). We analyze the OneStop Eye Movements corpus [4] and retain first readings from these two conditions. Repeated readings, question-screen events, answers, correctness, and all post-reading information are excluded.

OneStop was collected with participant consent under an institutional ethics protocol and released publicly after anonymization [4]. Our analysis uses only this public release.

The resulting data contain 19,438 trials from 360 readers and 30 articles, with nearly balanced goal labels. Each reader completed one goal condition, and the median trial contains 96 valid fixations. Reader identity, article identity, question content, session position, and condition metadata are not model inputs.

Each fixation is encoded by ten features: x and y coordinates normalized to the paragraph bounds; log fixation duration; horizontal and vertical displacement from the preceding fixation; log saccade amplitude; sine and cosine of the saccade angle; a line-transition indicator; and sequence position within the observed sequence. Retained fixations have positive finite durations and finite paragraph-screen coordinates. Continuous features are standardized with statistics estimated only from the current training set.

TABLE I
Model objectives, inputs, and trainable parameter counts.

Model	Objective	Model input	Parameters
Gaze-summary MLP	BCE	gaze summaries	11,265
Gaze GRU	BCE	fixation sequence	43,797
Text–gaze teacher	BCE	text + word-level gaze	1,038,945[a]
Text-only control	BCE	text	1,038,945[a]
Distilled GRU	BCE + KD	fixation sequence	43,797

[a] Trainable parameters; frozen RoBERTa is excluded.

B. Models

Table I summarizes the five neural model families. Models trained without distillation minimize binary cross-entropy (BCE). Gaze-summary MLP maps 22 sequence summaries to a 128–64 multilayer perceptron with ReLU and dropout 0.2. The summaries cover duration, displacement, saccade amplitude, line transitions, spatial dispersion, and path length.

Gaze GRU applies layer normalization to each fixation and processes the sequence with a two-layer unidirectional GRU [8] with hidden size 64. A 64–64–1 ReLU head with dropout 0.2 produces the logit. This model is the undistilled comparison for Distilled GRU. Sequences are zero-padded within each minibatch. The classifier uses the GRU output at the trial's final fixation before padding.

A frozen RoBERTa-base encoder produces subword representations [9]. Text–gaze teacher averages the representations belonging to each paragraph word. Eight word-level gaze features encode fixation count, total and mean dwell time, first-fixation and first-pass dwell time, regression-in and revisit counts, and skipping. Text and gaze features are projected to 128 and 64 dimensions, concatenated with a 32-dimensional learned word-position embedding, and mapped to width 192. A learned classification token and a three-layer, six-head Transformer [10] with feed-forward width 384, GELU, and dropout 0.1 produce the logit. A padding mask excludes padded word positions. Text-only control retains the text branch and fusion architecture but sets all gaze features to zero. It tests whether paragraph text alone predicts the reading goal.

Distilled GRU has exactly the same architecture and inputs as Gaze GRU. Let z_S and z_T be the student and teacher logits, and let $\tau = 2$. The student minimizes

$$\mathcal{L}_D = (1-\alpha)\,\mathrm{BCE}(y, \sigma(z_S)) + \alpha\tau^2\,\mathrm{KL}\left[q_T^{(\tau)} \,\middle\|\, q_S^{(\tau)}\right], \quad q_j^{(\tau)} = \mathrm{Bern}\left(\sigma\left(\frac{z_j}{\tau}\right)\right). \tag{1}$$

Teacher logits are generated only for student-training trials within each outer split. No held-out test trial is used to fit either model or construct a distillation target.

C. Evaluation Protocol

Readers are divided into five goal-stratified folds, and articles are divided into five folds. Their Cartesian product gives 25 disjoint test partitions that together cover all retained trials once. For each reader–article fold pair, training excludes every trial from either held-out fold, and the test set is their

intersection. All test readers and articles are therefore absent from training.

Within each outer training set, model selection uses three paired inner validation splits. Each validation split likewise holds out both its readers and articles from the corresponding inner training set. Learning rates are selected from $\{10^{-3}, 3\times 10^{-4}\}$ for the compact models and $\{3\times 10^{-4}, 10^{-4}\}$ for the teacher models. We also select $\alpha \in \{0.25, 0.50, 0.75\}$ for Distilled GRU. Balanced accuracy is the selection metric, with negative log-likelihood (NLL) as the first tie-breaker. Each final model is refitted for the median number of epochs selected across the three inner splits. The decision threshold is fixed at 0.5 before test evaluation. Test labels are not used for early stopping, calibration, or threshold selection.

Models are optimized with AdamW [11], weight decay 10^{-4}, and gradient clipping at 1.0. Inner training uses at most 30 epochs with a patience of five epochs. Batch sizes are 512 for Gaze-summary MLP, 256 for the GRU students, and 128 for the teacher models. Final refits use the selected learning rate and fixed epoch count described above, without test-dependent stopping.

D. Post Hoc Ablations

After the primary result, we ran post hoc controls using the same outer splits and training setup. Paired conditions inherit the selected learning rate and epoch count and keep the remaining student, teacher, or KD settings fixed.

The first control replaces teacher targets with binary label smoothing ($\epsilon = 0.10$) [12]. The second zeros every RoBERTa representation in an architecture-matched teacher while retaining its word-level gaze summaries. Across three fixed random seeds, we refit the BCE and label-smoothed GRUs, both teachers, and both KD students. Each student uses logits from the teacher trained with the same seed.

The third control cyclically reassigns teacher logits within each ground-truth class, preserving the class-conditional distributions while removing trial-level correspondence. It uses the same seeds and student settings.

We also fit logistic controls using either log dwell time per word [3] or the 22 gaze summaries. Their regularization is selected within the same nested reader/article splits.

E. Metrics and Uncertainty

Balanced accuracy (BA) is the primary metric because it averages the two goal-specific recalls [13]. Secondary metrics are AUROC, macro F1, negative log-likelihood (NLL), Brier score, and 15-bin expected calibration error (ECE) [14].

For the primary contrast between Distilled GRU and Gaze GRU, a paired 5,000-draw percentile bootstrap resamples whole readers within each goal. We report the central 95% interval. A 2,000-draw sensitivity analysis resamples readers within each goal and independently resamples articles, weighting each trial by the product of their multiplicities. Bootstrap samples are paired between models. We also report the number of outer partitions in which the primary contrast is positive. These partitions are not treated as independent trials. For each post hoc contrast, resampled units are paired across conditions, and the seed-specific BA differences are averaged within each draw.

TABLE II
POOLED OUT-OF-FOLD PERFORMANCE. LOWER IS BETTER FOR NLL, BRIER, AND ECE. LR CONTROLS WERE ADDED POST HOC.

Model	BA	AUROC	Macro F1	NLL	Brier	ECE
Dwell-time LR[b]	.5887	.6292	.5887	.6635	.2359	.0155
Gaze-summary LR[b]	.6066	.6493	.6065	.6572	.2327	.0236
Gaze-summary MLP	.5942	.6359	.5936	.6632	.2360	.0410
Gaze GRU	.5866	.6178	.5862	.6710	.2391	.0210
Text–gaze teacher	.6126	.6613	.6120	.6561	.2312	.0496
Text-only control	.4986	.5017	.4986	.6952	.2510	.0304
Distilled GRU	**.6205**	**.6708**	**.6204**	**.6427**	**.2265**	**.0142**

[b]Nested selection within the frozen reader/article splits.

F. CPU Measurement

For timing, we select the distilled development checkpoint whose full-sequence validation BA is closest to the mean across all inner evaluations. After 100 warm-up iterations, we time one full-sequence forward pass for each retained trial in randomized order. Measurements use an Intel Core Ultra 7 155H and one PyTorch thread, excluding input construction, parsing, and disk I/O.

IV. RESULTS

A. Confirmatory Performance

Table II reports pooled out-of-fold predictions. Distilled GRU improves BA over Gaze GRU by 0.0339. The reader-clustered 95% interval is $[0.0175, 0.0504]$, and the crossed reader/article interval is $[0.0130, 0.0553]$. Both exclude zero.

The improvement is positive in 18 of the 25 test partitions. Distilled GRU also exceeds Gaze-summary MLP and Text–gaze teacher in pooled BA, whereas Text-only control remains at chance. NLL, Brier score, and ECE are also lower than for Gaze GRU.

Among the post hoc logistic controls, Gaze-summary LR is strongest. Distilled GRU leads it by .0139 BA, with a crossed reader/article interval of $[-0.0037, 0.0321]$.

B. Post Hoc Ablations and Seed Sensitivity

Table III averages the post hoc comparisons over the matched random seeds. Label smoothing does not improve BA over the GRU. All KD conditions improve BA at every seed, with reader-clustered intervals above zero.

The two teachers yield similar students under both bootstrap schemes. Text–gaze KD exceeds class-shuffled KD at every seed, with a crossed interval of $[.0033, .0352]$. Class-shuffled KD remains above the GRU.

C. Inference Cost

Because distillation leaves the student unchanged, the distilled and undistilled GRUs have identical parameter counts and inference operations. The serialized student state occupies 176 KiB. Across all retained sequences, model-only latency has a 5.08-ms median and a 16.57-ms 95th percentile.

TABLE III
POST HOC MATCHED-SEED RESULTS. STUDENT Δ VALUES ARE RELATIVE TO GRU+BCE. DIRECT CONTRASTS FOLLOW THE MODEL ROWS. CIS USE READER-CLUSTERED RESAMPLING.

Condition	BA	ΔBA	Reader 95% CI	Seeds > 0
GRU+BCE	.5857	–	–	–
+ label smoothing	.5852	−.0005	[−.0022,.0011]	1/3
+ gaze-only KD	.6221	+.0364	[.0199,.0534]	3/3
+ text–gaze KD	.6232	+.0376	[.0223,.0529]	3/3
+ class-shuffled KD	.6042	+.0185	[.0052,.0317]	3/3
Text–gaze KD − gaze-only KD	–	+.0011	[−.0030,.0052]	1/3
Text–gaze KD − class-shuffled	–	+.0190	[.0061,.0317]	3/3

V. DISCUSSION

Teacher-derived soft targets improve the fixed gaze-only GRU without changing its inference path. The tested label-smoothing condition does not provide a comparable benefit. The gaze-summary logistic control narrows the margin, so the evidence supports distillation for this GRU rather than superiority over all gaze-only models.

Removing contextual text has no measurable effect on student performance. This does not support a text-specific benefit. Both teachers retain word-level gaze aggregation and greater capacity than the student, so the experiment does not separate these factors.

Within-class shuffling cuts the matched-seed gain roughly in half but does not remove it. Trial-level correspondence contributes to the gain. The remainder may reflect class-dependent soft targets or their optimization effect.

The ablations were designed after the primary result, and three seeds do not characterize optimization variability. The confidence intervals quantify reader and article sampling after averaging across those seeds. Because reading goal is assigned between readers, a crossover study is needed to identify within-reader effects. The data come from one English news corpus and a laboratory eye tracker, and all predictions use completed scanpaths. Other domains, trackers, self-selected goals, and partial sequences require separate evaluation.

The gaze-only student still requires calibrated coordinates and paragraph bounds. End-to-end device studies should include eye tracking, preprocessing, memory, and energy, which are outside the CPU benchmark. Any interface adaptation should be opt-in and evaluated with users.

VI. CONCLUSION

Under simultaneous reader and article holdout, teacher distillation improves a compact gaze-only classifier at unchanged inference cost. Within-class target shuffling reduces but does not eliminate the gain. Contextual text provides no measurable improvement in student performance.

CODE AVAILABILITY

Code, configurations, and aggregate results are available at github.com/gohyunsu/gaze-only-reading-goal-distillation.

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